Total Printed Pages: 02

Roll No.

PYB -101

Examination – June - 2023
B.Tech. I/II Sem : Common to all Branches
Applied Physics

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit to be observed is: Part a & b – Objective type/Fill. Part iii – Max 100 words Part iv – Max 400 words. Word limit NOT to be followed for diagram, numerical and derivations.

Q.1 (i) Which one of the following is not correct for photons: 02

(a) They possess energy (b) They possess mass

(c) They possess momentum (d) They possess intensity

(ii) Matter waves have mass 02

(a) True (b) False

(iii) Explain Heisenberg's uncertainty principle. 03

OR

Write the energy and momentum operators. 03

(iv) Write a brief note on phase and group velocities. 05

OR

Write a brief note on Schrödinger wave equation. 05

Q.2 (i) Ruby laser belongs to which of the following category: 02

(a) Atomic (b) Molecular (c) Ionic (d) Electronic

(ii) Optical fibers works on the principle of 02

(a) Interference (b) Diffraction (c) Dispersion (d) Total internal reflection

(iii) Explain briefly, how holograms are constructed. 03

OR

Enlist various important properties of a laser radiation. 03

(iv) Deduce the expression for intermodal dispersion in a step index optical fiber. 05

OR

Explain construction and working of a He-Ne laser. 05

Q.3 (i) The emitted wavelength λ from a semiconductor of energy band gap E_g is 02
(a) E_g/hc (b) hc/E_g (c) hv/E_g (d) E_g/hv

(ii) Can solar cell also be classified as a photodiode? 02
(a) Yes (b) No

(iii) Write a note on energy bands in a semiconductors. 03

OR

Explain working of a LED. 03

(iv) Explain the concept of effective mass in solid state physics. 05

OR

Write a note on working of solar cell. 05

Q.4 (i) The phenomenon of superconductivity is sensitive to magnetic field 02
(a) True (b) False

(ii) As we reduce the size, the surface area (per unit mass) 02
(a) Increases (B) Decreases

(iii) What are Cooper pairs in a superconductor? 03

Give a brief account of some applications of nanotechnology. 03

(iv) Differentiate between type I and Type II superconductors. 05

OR

Write a note on Carbon nanotubes. 05

Q.5 (i) Dielectric materials are 02
(a) Conductors (b) Semiconductors (c) Insulators (d) Superconductors

(ii) Piezoelectric effect can be direct or it can be indirect 02
(a) True (b) False

(iii) Differentiate between polar and non-polar dielectrics. 03

OR

Explain how the piezoelectric effect originates. 03

(iv) Establish relation between **E**, **D** and **P** vectors for a dielectric material. 05

OR

Write a note on various applications of piezoelectric materials. 05

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